

Linear Algebra — Exercises

Complex numbers

Exercise 1. Calculate: a) $(2+3i)(4-2i)$ b) $(1+2i)(2i-1)$ c) $(3+i)(3-i)$ d) $(a+bi)(a-bi)$

Exercise 2. Calculate: a) $\frac{2-3i}{3+4i}$ b) $\frac{i}{1+i}$ c) $\frac{3-4i}{i}$

Exercise 3. Calculate: a) $\frac{1+4i}{4-i}$ b) $\frac{(3+4i)^4}{(3-4i)^3}$

Exercise 4. Show that for every two complex numbers z and w :

a) $\overline{z+w} = \bar{z} + \bar{w}$ and $\overline{zw} = \bar{z} \bar{w}$

b) $|zw| = |z| |w|$

c) $\left| \frac{z}{w} \right| = \frac{|z|}{|w|}$

Exercise 5. Show that multiplication of complex numbers is associative.

Exercise 6. Show that multiplication of complex numbers is distributive with respect to addition.

Exercise 7. Describe geometrically the sets of points $z \in \mathbb{C}$ satisfying the following conditions:

a) $|z - i + 3| = 5$

e) $\Re((z-i)^2) = 0$

b) $|z - i + 3| > 5$

f) $\Re(z^2) = 2$ and $(\Im(z+i))^2 = 1$

c) $|z - 1 + 3i| \leq 5$

g) $\left| \frac{z+3}{z-2i} \right| \geq 1$

d) $2 \leq |z+i| < 4$

h) $\Re(z+1) < 0$ and $|i-z| \leq 3$

Exercise 8. Find polar forms of the following complex numbers:

a) $1+i$

g) $1-i\sqrt{3}$

b) -3 (negative 3)

h) $-z$, if $z = \cos \alpha + i \sin \alpha$

c) $-i$

i) $\sqrt{3} - i$

d) $2+2i$

j) $\bar{z}$, if $z = \cos \alpha + i \sin \alpha$

e) $3+3\sqrt{3}i$

k) $\frac{1}{2} + \frac{\sqrt{3}}{2}i$

f) $z = \sin \alpha + i \cos \alpha$

Exercise 9. Calculate:

a) $\sqrt[3]{(1+i)^6}$

d) $\left(\frac{2+2i}{\sqrt{3}-i} \right)^8$

b) $\left(\frac{1-i}{1+i} \right)^{222}$

e) $\sqrt[6]{1}$

c) $\left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)^{124}$

f) $\sqrt[4]{-1}$

g) $\sqrt[4]{(3-5i)^4}$

h) $\sqrt[6]{-1}$

Exercise 10. Describe geometrically the sets of all the following roots: a) $\sqrt[6]{1}$ b) $\sqrt[4]{-1}$ c) $\sqrt[8]{1}$
d) $\sqrt[6]{-1}$

Exercise 11. Use De Moivre's formula to:

- (a) express $\sin 5\alpha$ in terms of $\sin \alpha$ and $\cos \alpha$,
- (b) express $\cos 4\alpha$ in terms of $\sin \alpha$ and $\cos \alpha$.

Exercise 12. Prove that

$$\frac{\cos \alpha + i \sin \alpha}{\cos \beta + i \sin \beta} = \cos(\alpha - \beta) + i \sin(\alpha - \beta).$$

Exercise 13. Find the polar form of: a) $2 + \sqrt{2} + i\sqrt{2}$ b) $2 + i + i\sqrt{3}$

Exercise 14. Calculate: a) $\cos \frac{\pi}{12}$ b) $\cos 105^\circ$ c) $\cos 75^\circ$ d) $\cos(-15^\circ)$

Exercise 15. Calculate $(1 + \cos \alpha + i \sin \alpha)^{10}$ for $\alpha \in [0, \pi)$.

Exercise 16. Let $f(x) \in \mathbb{R}[x]$. Show that z is a root of f if and only if $\bar{z}$ is a root of f .

Exercise 17. Factor each of the following polynomials into a product of factors of degree at most 2:

- a) $x^6 + 1$ b) $x^4 + 1$ c) $x^6 + x^3 + 1$ d) $x^5 + 1$

Exercise 18. Prove that every polynomial of an odd degree, with all real coefficients, has at least one real root.

Exercise 19. Knowing that $1 - 2i$ is a root of the polynomial $z^4 + 2z^3 + 2z^2 + 10z + 25$, find all remaining roots.

Exercise 20. Knowing that $2 + i$ is a root of the polynomial $z^4 - 2z^3 - 6z^2 + 22z - 15$, find all remaining roots.

Exercise 21. Solve the equations: a) $x^2 + x + 1 = 0$ b) $2x^2 + ix + 3 = 0$ c) $x^2 + 2x - i\sqrt{3} = 0$